

University Hospitals of Leicester NHS Trust NHS				Actual Body Weight: _____ kg Aminophylline does not distribute fully into fat. If the patient is obese use ideal body weight (IBW) for all calculations Male: IBW 50kg + 2.3kg for each inch over 5ft tall Female: IBW 45.5kg + 2.3kg for each inch over 5ft tall Ideal Body Weight: _____ kg				Attach addressograph here			
Intravenous Infusion Administration Record Please attach to patient's prescription chart											
INTRAVENOUS LOADING DOSE = 5mg/kg over 20 minutes											
Do not give a loading dose if the patient is taking oral aminophylline or theophylline											
AMINOPHYLLINE		mg in 100ml sodium chloride 0.9% over 20mins		Date	Sign	Print	Bleep	Fluid Batch No		Sign	
								Start Time		Check	
INTRAVENOUS MAINTENANCE DOSE:				Patient	Dose	40kg	50kg	60kg	70kg	80kg	90kg
As a continuous infusion				Elderly/Heart Failure	0.3mg/kg/hr	12	15	18	21	24	27
All doses in ml/hr				Non-Smoking Adult	0.5mg/kg/hr	20	25	30	35	40	45
				Smoking Adult	0.7mg/kg/hr	28	35	42	49	56	63
Date	Aminophylline 500mg in 500ml Sodium Chloride 0.9% or Glucose 5% (delete as required)	Dose (mg/kg/hr)	Rate (ml/hr)	Sign	Print	Batch No.	Start Time	Sign	Witness	Pharm	Level
Date	Aminophylline 500mg in 500ml Sodium Chloride 0.9% or Glucose 5% (delete as required)	Dose (mg/kg/hr)	Rate (ml/hr)	Sign	Print	Batch No.	Start Time	Sign	Witness	Pharm	Level
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PRESCRIBING INFORMATION FOR THE ADMINISTRATION OF INTRAVENOUS AMINOPHYLLINE

Discuss with your pharmacist for any advice needed

1. Measuring Theophylline Levels

The usual therapeutic range for theophylline is 10-20mg/L

Theophylline (not aminophylline) is measured as that is the active constituent of the drug

When should levels be checked?

If a loading dose has been given: 4-6 hours after starting infusion

If toxicity is a risk or suspected: 4-6 hours after starting infusion

All other patients: 18 hours after starting infusion

Repeat levels should be checked every 24 hours while on infusion

Adjustment to infusion rates can be calculated using the equation:

$$\text{New Infusion Rate} = \frac{15 \times \text{Current Infusion Rate}}{\text{Current Theophylline Level (mg/L)}}$$

2. Converting from IV to Oral

Not all patients requiring intravenous aminophylline need to be started on long-term oral aminophylline or theophylline

When converting from aminophylline to **oral theophylline**, a conversion factor is necessary. Aminophylline contains 80% theophylline therefore the conversion factor = 0.8. Multiply the total amount of aminophylline received in the previous 24 hours by 0.8 to obtain the total daily dose of theophylline. This should be given as two equal doses given twelve hours apart.

$$\text{Daily Dose of Oral Theophylline} = \text{Hourly Infusion Rate} \times 24 \times 0.8$$

If converting to **oral aminophylline** no conversion factor is needed. Simply multiply the hourly infusion rate by 24 and give as two equal doses twelve hours apart. Give the first oral dose as the infusion is reduced to zero over 8-12 hours. Take a theophylline level TWO days after converting to oral therapy

Oral theophylline is available as Uniphyllin MR (200mg, 300mg and 400mg tablets)
Oral aminophylline is available as Phyllocontin MR (225mg and 350mg tablets)

3. Interactions

Aminophylline is hepatically metabolised and the clearance of aminophylline is affected by other medication:

Clearance may be REDUCED (theophylline level increased) by:
erythromycin, clarithromycin, ciprofloxacin, diltiazem, verapamil, fluconazole, cimetidine, zafirlukast

Clearance may be INCREASED (theophylline level decreased) by:
lithium, phenytoin, rifampicin, carbamazepine, phenobarbitone, smoking

Consult the ward (or on-call) pharmacist for further advice

4. See the 'Aminophylline: procedure for the routine administration in adults' Monograph on the DMS (Document ID 10943)